Student Proposed?	Y/N	N/A
ID	**RAV2026-07**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**Dr MK Mason (MMRU, IDM)**	
Title	**Compact sealed stage-top incubator for long-duration live bacterial imaging**	

Description

Background

Mycobacteria are not uniform populations of simple rod-shaped cells. Even within a genetically identical culture, individual cells differ in growth rate, division timing, morphology, stress response and antibiotic susceptibility (**Aldridge** et al., 2012). These differences matter because mycobacterial behaviour is often expressed at the single-cell and microcolony scale: cells elongate, divide asymmetrically, form cords and organised aggregates, and interact with surfaces in ways that are lost in bulk culture measurements. Understanding these processes requires live imaging over time, at sufficient resolution to follow individual cells and the structures they form.

This need sits within a broader UCT mycobacterial imaging programme, where microscopy is used to study the physical and biological behaviour of mycobacteria, including cording, surface-associated growth and host-relevant infection models. In the longer term, this work links to contained imaging approaches for pathogenic mycobacteria and lung-relevant systems, including cell monolayers, Matrigel-supported interfaces and precision-cut lung slices. However, before such workflows can be translated into more specialised containment settings, the underlying imaging hardware must be robust, well characterised and safe to develop. A practical engineering step is therefore to build and validate the environmental control platform using a non-pathogenic model organism.

Long-duration time-lapse microscopy places strict demands on the sample environment. Temperature affects growth rate and division timing, small agarose-pad or chamber volumes are vulnerable to evaporation, and changes in humidity, gas composition or thermal gradients can alter the biology being measured (**Frigault** et al., 2009). The sample must remain on the microscope for hours to days, which means that incubation has to happen at the microscope stage itself. The system must maintain a stable environment while preserving optical access, avoiding condensation, and limiting focal-plane drift. If these conditions are not controlled, the resulting images may reflect stress caused by the imaging setup rather than the biological process under investigation.

Commercial stage-top incubators can provide temperature, humidity and gas control, but they are expensive, closed to modification, and usually designed for general mammalian cell culture rather than adaptable bacterial live-imaging workflows (**O'Carroll** et al., 2023; **Yan** et al., 2024). For this collaboration, the useful system is a compact, modifiable and well-characterised platform that can be built, repaired and adapted locally. The engineering challenge is therefore to design more than a heated box: the system must regulate temperature, humidity and carbon dioxide in a small sealed chamber, remain compatible with an inverted microscope, and provide stable imaging conditions for biologically meaningful bacterial time-lapse experiments.

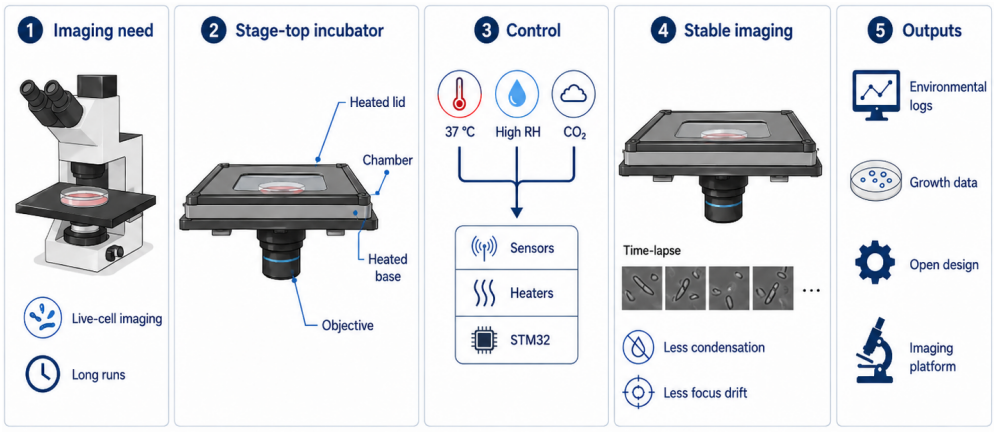


Fig. 7. Proposed stage-top incubator for long-duration bacterial imaging. A compact heated chamber maintains temperature, humidity and CO_2 around the sample while allowing time-lapse imaging on an inverted microscope. Schematic generated with assistance from ChatGPT (OpenAI, 2026) and edited by the author.

Subject of investigation

This project designs, builds and evaluates a compact sealed stage-top incubator for long-duration bacterial time-lapse imaging on an inverted microscope. The project will be developed as a safe engineering prototype using *Mycobacterium smegmatis*, a fast-growing, non-pathogenic mycobacterium that provides a useful model for testing live-imaging conditions, cell growth, morphology and microcolony behaviour without the containment requirements of pathogenic *M. tuberculosis* (**Sparks** et al., 2023; **Joyce** et al., 2011; **Aldridge** et al., 2012).

The incubator must maintain a stable internal environment near 37°C, with high humidity to reduce evaporation from small imaging volumes and controlled carbon dioxide where required. It must be compact enough to mount on a standard microscope stage, preserve optical access for time-lapse imaging, and address practical failure modes including condensation, uneven heating, thermal settling and focal-plane drift.

The internal conditions will be measured using digital temperature, humidity and carbon dioxide sensors, with an STM32-based controller used to drive the heaters and gas/humidity actuators, implement closed-loop control, and log the environmental data. The candidate will characterise the incubator in terms of setpoint tracking, stability, uniformity, repeatability, long-duration drift, condensation behaviour, image quality and focal stability.

The final system will be demonstrated on the microscope using live *M. smegmatis*, in consultation with the co-supervisor. The aim is to produce a working, characterised and modifiable stage-top incubator that supports current bacterial imaging while creating an engineering foundation for future mycobacterial imaging work, including cording, surface-associated growth and lung-relevant infection models, while keeping the student project within a safe development and validation scope.

Deliverables	**The key objectives of this project are:** (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of previously published research in this field and critically evaluate current technologies and research that applies to this project. (c) To specify the operating point (a temperature near 37°C, a controlled near-saturated humidity and a defined carbon dioxide atmosphere) and the optical and mechanical constraints for mounting on a standard inverted microscope, and to select appropriate sensors and actuators. (d) To design and build the sealed chamber with heated optical windows, humidified gas path, and CO_2 mixing, within the footprint of a microscope stage. (e) To develop and implement closed-loop control and data-logging algorithms for the coupled multivariable system on an embedded platform. (f) To test and evaluate the system performance, characterising temperature stability and focus drift, condensation, carbon dioxide and humidity tracking, and uniformity across the imaging field, and to demonstrate imaging of a live *M. smegmatis* sample on the microscope. (g) To discuss the performance of the system, draw conclusions, and make recommendations for future improvements. **The key deliverables of this project are:** (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) A working prototype stage-top incubator that mounts on a standard inverted microscope and holds near 37°C, a controlled humidity and a defined carbon dioxide atmosphere under closed-loop control, with a host-side logging interface. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Control systems, Embedded programming, Interest in instrumentation, optics and microbiology
Extra Information	**References** B. B. **Aldridge**, M. Fernandez-Suarez, D. Heller, V. Ambravaneswaran, D. Irimia, M. Toner, and S. M. Fortune. Asymmetry and aging of mycobacterial cells lead to variable growth and antibiotic susceptibility. *Science*, 335(6064): 100–104, 2012. **doi:** 10.1126/science.1216166 M. M. **Frigault**, J. Lacoste, J. L. Swift, and C. M. Brown. Live-cell microscopy — tips and tools. *Journal of Cell Science*, 122(6):753–767, 2009. **doi:** 10.1242/jcs.033837 G. **Joyce**, B. D. Robertson, and K. J. Williams. A modified agar pad method for mycobacterial live-cell imaging. *BMC Research Notes*, 4:73, 2011. **doi:** 10.1186/1756-0500-4-73 M. K. **Mason** and R. A. Verrinder. Biocontainment for live imaging of mycobacterial droplet evaporation: Uct Block Grant feasibility work. Molecular Mycobacteriology Research Unit, Institute of Infectious Disease and Molecular Medicine, UCT, 2024. Collaborative feasibility study R. **O'Carroll**, J. P. Reynolds, M. Al-Roqi, E. D. Aiyegbusi, and D. Dooley. ThermoCyte: an inexpensive open-source temperature control system for *in vitro* live-cell imaging. *Royal Society Open Science*, 10(11):231037, 2023. **doi:** 10.1098/rsos.231037 I. L. **Sparks**, K. M. Derbyshire, W. R. Jacobs, and Y. S. Morita. *Mycobacterium smegmatis*: the vanguard of mycobacterial research. *Journal of Bacteriology*, 205(1):e00337–22, 2023. **doi:** 10.1128/jb.00337-22 H. **Yan**, T. Wu, X. Li, Z. Feng, M. Ge, L. Zhang, and W.-F. Dong. Establishment of the microscope incubation system and its application in evaluating tumour treatment effects through real-time live cellular imaging. *Frontiers in Bioengineering and Biotechnology*, 12:1447265, 2024. **doi:** 10.3389/fbioe.2024.1447265 **Note:** the biological samples (*Mycobacterium smegmatis*, a non-pathogenic BSL-1 model organism) and the imaging protocol will be arranged with the co-supervisor, and any sterility or containment requirements will be handled through the co-supervisor's unit (MMRU, IDM).
Broad Research Area	Environmental control, Mechatronics, Embedded Systems, Biomedical instrumentation, Live-cell imaging
Project suitable for: ME/ECE/EE/ALL?	EE, EC, ME

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EEE4022S Topic RAV2026-07 GAs

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GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the design and control of a sealed multivariable environment that must also function as part of an optical instrument. It involves ill-posed and under-specified problems, such as the: • specification of an achievable operating point and of acceptable stability, uniformity and focus-drift tolerances for imaging, • design of a controller for three coupled variables, temperature, humidity and CO_2, regulated by actuators with different time constants. It also presents high-level problems, such as: • Reconciling conflicting requirements, in particular a gas-tight seal to hold a defined carbon dioxide atmosphere against the objective access that imaging needs, and heated optical windows against the optical clarity and low condensation that imaging demands. • Achieving temperature stability tight enough to keep focus drift within micrometres over long imaging runs. • Managing the cross-coupling between the temperature, humidity and carbon dioxide loops, where actuating one variable disturbs the others. • Working within the footprint, mass and safety constraints of a microscope stage. The project also necessitates handling complex and incomplete information by: • Synthesising knowledge from thermal design, gas handling, optics, control theory and embedded systems, alongside the imaging requirements set by the co-supervisor, where no single standard solution exists. • Balancing technical performance, build cost and the needs of the researchers who will run imaging experiments in the chamber.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative investigation and experimentation to: • Identify the dynamic response of each subsystem, measuring step responses and the cross-coupling between the temperature, humidity and carbon dioxide loops. • Design and conduct experiments to quantify focus drift and condensation on the optics, and to tune and validate the closed-loop controllers against defined setpoints. • Map the uniformity of the internal environment across the imaging field. • Apply statistical analysis to quantify stability, repeatability and steady-state error, and to demonstrate a representative imaging run on the microscope.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be both computational and practical in nature and will require competencies in embedded programming, control design, data handling and analysis, and mechanical, thermal and electronic design. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate tools to achieve the project objectives. **Software and tools required may include:** • C for embedded programming of the STM32 firmware, using the HAL • MATLAB/Python for control design, system identification and simulation • MATLAB/Python for data processing, analysis and validation, including image-based quantification of focus drift • CAD for the stage-top mechanics, optical mounts and the gas and moisture handling paths • Circuit design tools for the heater, valve and sensor interface electronics	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.